Particles or waves?
Research shows that antimatter, just like matter might behave like both particles and waves **https://dgit.in/june19-46**

Time traveling particle
Physicists are looking at simulations to find particles that can potentially jump back in time **https://dgit.in/june19-47**

models utilised to study and state the relationship between variables, in this case, the target and the potential chemical to cure it. These models are known as Quantitative structure-activity regression (QSAR) models. In this particular step, alterations are made to the chemical structure of identified chemicals to improve their efficiency and the overall safety of the drug while minimising all the potential side-effects.

Research firm McKinsey estimates that machine learning and big data in pharmaceuticals and medicine could possibly generate a value of up to $100 billion every year. The firm believes that ML helps in reducing costs and timeframes for drug discovery. They also foresee that the predictive modelling of biological processes and drugs, accomplished by ML, could possibly help in finding new potential candidate molecules which sport a high probability of being successfully developed. Costly hurdles such as delays and any other disadvantageous events could be avoided by real-time monitoring of molecules and chemicals, and removal of data silos.

USAGE OF ML IN IMAGE ANALYSIS

High content screening of cellular phenotypes is a key tool that supports early drug discovery. The term 'high content' actually pertains to the rich set of thousands of predefined features like shape, size or texture. These features are extracted from pictures using classical image processing methods. High content screening allows examining of microscopic images which provide aid in the study of the effects of multiple chemical treatments on different cell cultures.

Machine learning offers a solid promise in this area since relevant image features that can differentiate one treatment from another are 'automatically' grasped from the data. ML, coupled with deep neural network (DNN) acceleration can be applied by biologists and data scientists who hope to speed up the analysis of high content imaging screens. DNN acceleration techniques can process multiple images in significantly lesser time than traditional methods of image analysis in drug discovery and development. They also extract a greater amount of information and insight from image feature which the model finally ends up learning.

LIMITATIONS OF ML IN DRUG DISCOVERY

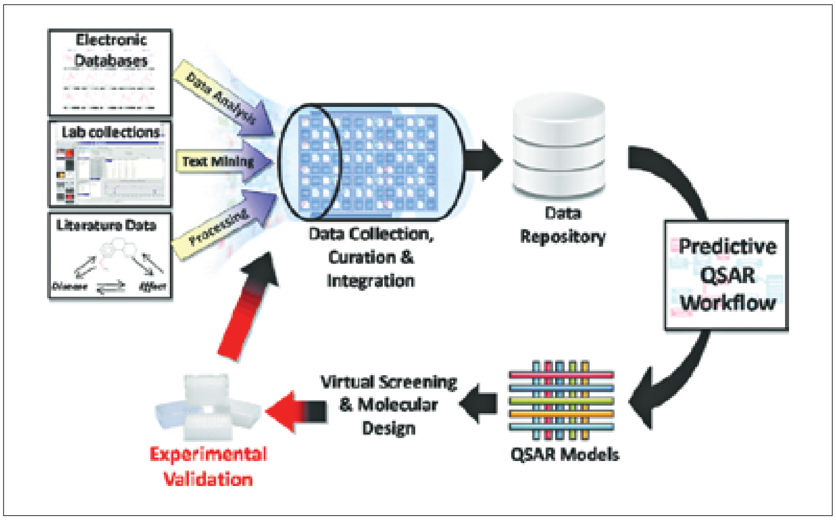
There are certain challenges when it comes to the application of machine learning in drug discovery. Despite there being no lack of data to collect, the complex nature of the data and the fact that it is often siloed is what researchers in this field must combat. Progress in drug development with ML is heavily reliant on the ability to efficiently access, amalgamate and use the complex medical data from clinical trials - both internal and external. The sheer scale of this - in terms of variance of data standards and the wide range of data of different types being collected - make it extremely time-consuming and costly to standardise. However, once standardised, this method can reduce the costs of drug discovery and development significantly, as suggested in the sections above.

When it comes to image analysis and classification, machine learning methods largely depend on huge expert-labelled datasets to train their models systematically. The amount of time and manual effort required to make such extensive datasets is often a roadblock that can seem insurmountable.

Machine learning can accelerate high content screening

Another issue presents itself in the clinical trial stage of drug discovery and development. Although the use of machine learning is highly beneficial in the nascent stages of drug discovery, ML cannot be used much beyond the hypothesising phase since there is no real substitute for clinical trials which require actual humans.

However, machine learning can be extremely useful in the aftermath of clinical trials since it can be used to collect and store all the data collected from the trial for easy analysis and recordkeeping. Researchers are only now beginning to understand the broader applicability and potential of using machine learning in drug discovery and development beyond simply applying it in structure-property regressions.



The QSAR models used in drug discovery could be enhanced by using ML.